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Full Length Article

VGM-UNet: A hybrid visual graph deformable mamba with fourier neural operator U-Net for medical image segmentation

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ABSTRACT

Deep learning methods have demonstrated remarkable advancements in medical image segmentation. However, achieving high accuracy remains a prominent challenge. In this paper, we present a new architecture, named VGM-UNet, that improves U-shaped segmentation models in performance and expressiveness by introducing the Structured State Space Duality algorithm to combine Graph Neural Networks, sparse attention, and Mamba-2, into U-Net and yield the best of these designs. This is accomplished through three primary modifications: we first adopt a novel approach, constructing a 2D State Space Model and an eight-way multi-scanning module, thereby creating the Vision Mamba-2, which serves as the foundation for building a hierarchical visual backbone that can be directly applied to the graph structure of image patches. Then, based on the Fast Fourier transform, we construct a Feed-Forward Network module, as a complement to Mamba, to model channel contents and improve the accuracy of capturing small objects. Moreover, on the basis of the modular architecture, we build a simple yet powerful U-shaped hybrid network, which simplifies the model design and enhances the model's expressiveness. These changes alleviate the limitations of conventional U-shaped architectures in accuracy improvements and achieve impressive results. We validate VGM-UNet through extensive experiments, demonstrating that our model outperforms existing state-of-the-art models in terms of segmentation accuracy on the Synapse and ACDC benchmark datasets. The experimental results also indicate that the Visual Graph State Space module can be conveniently applied to various medical image segmentation tasks.

1. Introduction

Computer-assisted medical image analysis system is one of the essential tools that can facilitate medical professionals to make evaluations during diagnosis, treatment, and continuous monitoring of patients' health conditions (Alam et al., 2023). The accuracy of the anatomical and pathological knowledge derived from medical images can affect the quality of medical practice to a great extent (Xia et al., 2025). Due to the high resolution of the images and the presence of numerous interrelated fine structures within them, powerful methods that can precisely express global features to locate lesion areas and capture as many local features as possible to distinguish ambiguous boundaries and subtle details are highly desired.

Recent years have witnessed vast developments of deep learning models in medical image segmentation. Particularly, Convolutional Neural Networks (CNNs) (Lecun et al., 1998) and Vision Transformer (ViT) (Dosovitskiy et al., 2020) based models possess the capability of obtaining hierarchical feature representations with input feature maps, thus making tremendous contributions to this field. Despite their success, the conventional architectures still face challenges in achieving high accuracy due to their inherent limitations. First, medical images typically contain irregular and complex objects, and the distribution of regions of interest within them is also uneven, but the commonly used grid of CNNs or the sequence structures of ViT are inflexible to capture these irregular-shaped and complex-structured objects. Second, CNNs lack the ability to model long-range relationships as they primarily focus on capturing local features, which limits their potential for improvements in precision. On the other hand, transformers excel in extracting long-term relationships, but their quadratic computational complexity of self-attention with respect to the sequence length of image size leads to high computation costs, and their performance saturates fast when scaling deeper (Wang et al., 2022a), which makes them resource-intensive and limits their practical use.

The recent work Mamba-1 (Gu & Mamba, 2023), a novel State Space Model (SSM) (Nguyen et al., 2022), integrating time-varying parameters to selectively filter out key information in an input dependent manner and combining a hardware-aware algorithm for concurrent computation to facilitate the training and inference processes, has emerged as a highly

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promising approach for long sequence modeling with linear complexity. Research on Mamba has demonstrated its potential as an alternative to Transformers in natural language processing tasks. However, Mamba is designed to process data in a specific direction, where the preceding data are not connected with the subsequent data, which affects its adaptability to dealing with visual image data. Although subsequent efforts attempt to introduce desirable properties of vision models into Mamba-1, the core algorithm relies on the hardware-aware parallelism. It is still less efficient than hardware-friendly models because it does not utilize matrix multiplication units, while modern accelerators are specifically designed for this purpose. For instance, VMamba (Liu et al., 2024c) proposes a two-dimensional selective scan (SS2D), which performs spatial domain traversal by conducting selective scanning with SSM in both horizontal and vertical directions, thereby extracting relationships and obtaining contextual knowledge through the compressed hidden states calculated along the corresponding scanning paths. Many studies (Chen et al., 2025; Ma et al., 2024; Wang et al., 2024b) introduce SS2D as the basis to construct U-shaped networks for medical image tasks, with the core inherited from Mamba-1, and have difficulty achieving a high performance.

The aforementioned limitations in medical image segmentation tasks underscore the need for innovative, hardware-friendly solutions that can effectively handle high-resolution images with interrelated delicate structures, which can obtain a large amount of contextual information and model both localized fine-grained features and long-range dependencies in medical images to improve segmentation accuracy. In this paper, we draw inspiration from the structured state space duality (SSD) framework of Mamba-2 (Dao & Gu, 2024), and suggest that Vision Mamba-2, an adaptation of Mamba-2 (Dao & Gu, 2024) merging state space principles with selective attention mechanisms, can be strategically connected to the visual attention to improve performance and robustness. The upgraded Mamba-2 (Dao & Gu, 2024) incorporates the Structured State Space Duality (SSD) algorithm and a variety of advanced techniques, such as multi-head State Space Models (SSMs) (Nguyen et al., 2022), tensor parallelism, and sequence parallelism, and has the potential to be an effective approach to complex visual tasks. However, we notice challenges when directly applying Mamba-2 (Dao & Gu, 2024) for medical image semantic segmentation due to the following reasons: 1) Mamba-2 (Dao & Gu, 2024) inherently lacks a sequential arrangement of visual components. 2) The stability issue of Mamba when scaling to large networks for vision datasets. 3) Recent SSMs have not addressed channel modeling explicitly. These obstacles render the task of constructing an effective visual representation learning model nontrivial. To this end, we propose Visual Graph Mamba-2 (VGM), a vision backbone introducing Mamba-2 (Dao & Gu, 2024) into Transformers to leverage the strengths of both to facilitate effective visual representation learning, which serves as the basis for constructing the architecture of VGM-UNet to learn the contextual knowledge from medical images for semantic segmentation. Our core idea lies in integrating the SSD algorithm into the U-Net model, combining graph neural networks, sparse attention mechanisms, and Mamba-2 (Dao & Gu, 2024), and optimizing the performance and expressive power of the U-shaped segmentation model. This allows us to fully leverage these advantages and achieve the best results of these designs.

Our proposed approach enjoys three key technical contributions. First, to avoid the scaling quadratic complexity and ensure the model can capture irregular and complex objects in medical images, we develop the Visual Graph State Space Duality module. This module optimizes feature representation from three aspects: 1) Instead of treating an image as sequential patches or grids, we introduce graph representation learning to represent the image as a graph and use graph convolutional networks to extract the relationships between the graph nodes. 2) To reduce the situation where relevant features are missed due to the data-agnostic nature of handcrafted attention, we adopt a data-dependent sparse attention mechanism to select the desired features. 3) We leverage the advantages of Mamba-2 (Dao & Gu, 2024) to make the model flexible for handling long-range dependencies. Second, as a complement to Mamba, we construct a Feed-Forward Network module based on the Fast Fourier transform (FFT) to alleviate the limitations of Mamba through three optimization measures. 1) We apply a non-linear activation function to handle stability when extending the model to a large scale. 2) We utilize Rayleigh's Theorem on energy balance between the spatial domain and the frequency domain to enable our model to accurately capture small objects. 3) We employ the frequency convolution calculation method to mix the features from each channel, so as to make up for the deficiency of Mamba in extracting channel features. Third, unlike those image segmentation models built based on pure Mamba, we design a simple but powerful hybrid U-shaped network on a modular architecture. This network combines the advantages of graph representation, sparse attention, and SSMs. With this novel design, our model is not only capable of capturing short-term and long-term dependencies, but also selectively captures more informative features.

We empirically validate the effectiveness of our model on two benchmark datasets, Automated Cardiac Diagnosis Challenge (ACDC) (Zhang, 2024) and Synapse (Landman et al., 2015). Experiment results show our model outperforms the state-of-the-art models based on Transformers, Mamba-1 models and their variants. Finally, we demonstrate that our novel approach enables Mamba-2 to process 2D image data, the proposed network can flexibly scale to deeper layers, and this architecture can be easily integrated into various segmentation frameworks.

2. Related work

2.1. CNN-based medical image segmentation

CNNs are considered a substantial breakthrough and have become the mainstream deep learning technique for vision tasks. Nevertheless, the limitation in capturing the global receptive fields sets constraints on CNNs' effectiveness, and has led to innovation of diverse CNN variants, including those mainly using the U-shaped architecture for image segmentation. Recently, some researchers have adopted methods of integrating large kernel convolutions into the model, attempting to make it more able to capture global context information while constraining the computation cost in a manageable way, such as ConvUNeXt (Han et al., 2022) and CMUNEXT (Tang et al., 2024). MIU-Net (Xiao et al., 2025) incorporates multiple enhanced feature extraction functions, such as multi-scale feature extraction, a dual-branch attention capturing channel, and spatial relationships, thereby improving the segmentation of Organ-at-Risk. Another example is MSC U-Net (Zhang et al., 2025), in the encoder layer of which the input image is transformed into a pyramid structure, enabling the extraction of features at different scales, so as to improve its performance.

Additionally, some researchers have sought to enhance CNN's modeling capabilities to capture desirable features by leveraging attention mechanisms to focus on specific areas and objects in an image. For instance, EMCAD (Rahman et al., 2024) introduces mixing convolutional attention mechanisms to enhance feature extraction and constructs the decoder using multi-scale depth-wise convolution. AMSUnet (Yin et al., 2023) utilizes multi-scale and atrous convolution and a residual mechanism to improve feature extraction and feature fusion modeling ability, respectively. DRA-UNet (Wang et al., 2025) introduces an efficient channel attention and a triple attention, which are respectively used to enhance the connections between different stages in the encoder and decoder.

2.2. Transformers for medical image segmentation

Some researchers have introduced pure Transformers to segment images, but they encounter the challenge of quadratic scaling with increasing sequence length, which limits the applicability of this technique. Efforts have been made to mitigate the impact of this challenge. For example, CSWin Transformer (Dong et al., 2022) introduces a cross-shaped window to parallelly calculate the self-attention in horizontal and vertical stripes so as to constrain the computation cost. This successful model is fused into the U-shaped architecture of CSWin-UNet (Liu et al., 2025a) to facilitate the extraction of the interactions within input maps. UDTrans-Net (Wang et al., 2024a) notices that simple skip-conections in the U-Shaped architecture are less capable of capturing multi-scale features. The authors thus introduce a channel-wise and spatial-wise fusion attention to capture the channel and spatial relationships within the input features and a decoder-guided recalibration attention to mitigate the discrepancy between the features generated for the decoder.

In addition to pure Transformer-based segmentation models, some researchers have attempted to build hybrid architectures to capture local and global features for dense prediction. For instance, the first fully convolutional Transformer (FCT) (Tragakis et al., 2022) is proposed to address the fine-grained nature of medical images by performing a two-stage feature extraction through Convolutional Transformer modules. To obtain as much feature data as possible, MIST (Rahman et al., 2023) merges the merits of CNNs and Transformers, encodes the feature representation through multi-axis ViT, and segments the image through mixed convolutional attention. CT-Net (Zhang et al., 2024b) proposes an asymmetric asynchronous branch parallel structure by introducing and combining properties from both non-symmetric CNN and Transformer to capture the local and global relationships of the input features, and then fusing the low-coupling features together. Additionally, Zhu et al. (2023) employs the Swin Transformer as the backbone network to extract semantic feature information and construct a multi-model segmentation framework that integrates edge feature detection and feature fusion modules to enhance the accuracy of brain tumor segmentation tasks. However, this method, due to its adoption of a data-agonistic attention model, tends to overlook the useful features in other areas, and the approach of treating the image as a sequence of patches may lack flexibility when dealing with the diversity of brain tumors in MRI.

Furthermore, a novel method that aims to transfer zero-shot learning to new image segmentation tasks with well-designed and pre-trained prompts is gaining attention and has shown impressive performance in many segmentation competitions. Following this path, Sam2-UNet (Xiong et al., 2024) builds the encoder by introducing the Hiera backbone of SAM2 (Ravi et al., 2024), coupled with building the decoder using a classic U-shaped design. This model is more effective than the original SAM model. In addition, 3D-PMGNet (Zhu et al., 2025) achieves more accurate segmentation results by leveraging encoder features to generate and dynamically optimize probability maps. It also fuses these probability maps with up-sampled features and guides up-sampled features to give priority to the semantic regions with higher response rates. In summary, although many studies have attempted to introduce the desirable properties of CNNs into Transformers, such as capturing the local structure of the image through their local receptive fields, shared weights, and spatial subsampling, to enhance the expressiveness of their models, the instability in training and the over-smoothing between different blocks increase as the model deepens, which notably affect their segmentation outcomes.

2.3. Mamba variants for medical image segmentation

A novel approach, Mamba (Gu & Mamba, 2023), has emerged recently. It linearizes remote sequence complexity and trains efficiently through hardware-perceived parallelism, showing impressive performance in natural language processing. Specifically, Mamba integrates a selective mechanism to enhance the structured SSMs (S4) (Gu et al., 2021a, 2022, 2021b; Song et al., 2024), using a forget gate and selectively propagating methods to perform content-based reasoning. While effective in language processing, adapting the state-space language models for medical image processing is a non-trivial task. Several works have explored the potential of SSMs in visual tasks. For instance, VMamba (Liu et al., 2024c) adapts the Mamba language model to the design of an SS2D block as the core of a versatile visual backbone, which enables the comprehensive extraction of image features and has been adopted in various studies. The pioneer Vision Mamba UNet (VM-UNet) (Zhang et al., 2024a) builds an encoder and a decoder with SS2D blocks, with skip addition superseded by skip concatenation. Mamba-UNet (Wang et al., 2024b) introduces SS2D to build a simple U-shaped network, along with skip connections to preserve shallow spatial features.

Furthermore, some studies have incorporated tailored methods into the U-shaped architecture. Swin-UMamba (Liu et al., 2024a) effectively utilizes the capabilities of the pre-trained visual Mamba model and integrates it into the encoding process, while Swin-UMamba†(Liu et al., 2024b) integrates it into both the encoding and decoding processes, thereby constructing a U-shaped architecture to enhance the modeling ability for long-range dependencies in medical images. UM-Mamba (Chen et al., 2025) employs a unique 2D scanning approach, a four-way spiral selective scan, as the basis of the state space module in a U-shaped medical image segmentation network, enabling Mamba to learn circular-shaped features. SK-VM + + (Wu et al., 2025) constructs a parallel vision mamba layer and integrates it into the skip-connections of U-Net + +, making the skip-connections more effective, which differentiates them from the traditional U-shaped architecture, thereby enhancing the fusion of high and low feature information. Inheriting the adaptation features from nnU-Net (Isensee et al., 2018), U-Mamba (Ma et al., 2024) incorporates CNNs and SSMs as blocks to a hybrid U-shaped architecture for extracting local fine-grained features and capturing remote dependencies.

Moreover, some researchers have extended the State Space Model to process 3D medical image analysis. MMambaDiff (Liu et al., 2025b) proposes a semantic hierarchical embedding (SHE) mechanism and a global slice-aware Mamba (GSPM) layer to preserve the fine features and structural consistency of the segmentation results, thereby attempting to enhance the effectiveness of diffusion models in 3D medical image segmentation. To evaluate the effectiveness of Mamba in 3D medical image segmentation tasks compared with CNNs and transformers, Luca et al. (2025) constructs a bidirectional 3D Mamba layer as the foundation and integrate it into a U-shaped architecture to build various structural networks, such as SegMamba, BiSegMamba, and MultiSegMamba. However, Mamba-1 has some limitations restricting its practical application and its potential in strong segmentation performance. For instance, it lacks a theoretical framework that can connect with various attention variants, and the hardware-aware selective scanning mechanism is less efficient.

3. Methods

3.1. Hybrid U-Net

The architecture of Visual Graph Mamba UNet in the tiny version (VGM-UNet-T) is illustrated in Fig. 1. We introduce the SSD framework to construct the Visual Graph Mamba (VGM) block, as shown in Fig. 2(a), which serves as the basis for building a U-shaped network based on our modular architecture, with different numbers of blocks according to different scales. The design of this block inherits the SSD algorithm and combines graph representation learning, refined sparse attention, and SSMs for handling the graph structure of image patches. Thus, it can not only obtain short-term and long-range dependencies but also selectively capture desirable features. Next, as a complement to Mamba, we construct a Feed-Forward Network module based on the FFT by mixing the characteristics of the channels via the convolution operations in the frequency domain, to capture smaller objects according to the energy equivalence between image patches in both spatial and frequency domains, as stated in Rayleigh's Theorem.

Then, we construct a hybrid U-shaped segmentation network including an encoder, a decoder, and skip connections. Similar to Wang et al. (2022b), an input image with the dimension of $H \times W \times 3$ is partitioned into N tokenized patches, and then converted to an eigenvector as a set of unordered vertexes $V = \{v_1, v_2, \dots, v_N\}$. VGM blocks are applied

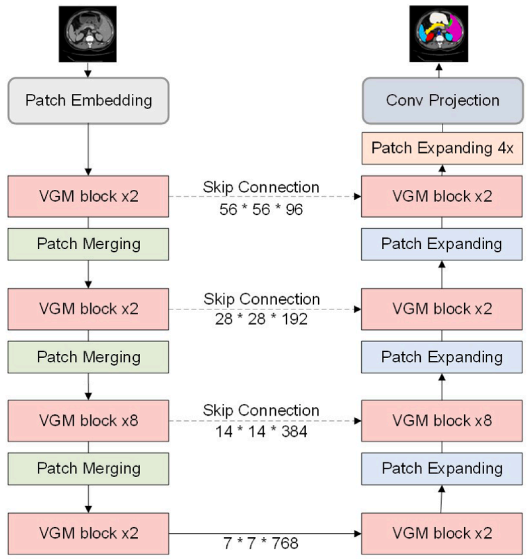

Fig. 1. Overview of the proposed VGM-UNet architecture with the encoder and decoder, both of which consist of four stages built on a novel Visual Graph Mamba (VGM) block.

afterward for feature transformation. Both the encoder and decoder function across four stages with a configurable number of VGM blocks. In the encoder, each of the first three stages consists of a VGM block stack and a patch merging layer represented as down-sampling, while the last stage only includes VGM blocks. During feature extraction, the patch merging layer halves the feature resolution and doubles the number of channels, generating a hierarchical feature representation simultaneously. The decoder involves a VGM block stack and a patch expanding layer for up-sampling at the first three stages, and a VGM block stack at the last stage. Corresponding to the patch merging layer utilized in the encoder, the patch expanding layer is an up-sampling of the extracted feature map to identify the regions of interest and double the feature resolution. The skip connections are utilized to retain the missing spatial features caused by down-sampling and to concatenate features from the corresponding encoder layers at the decoder layers to boost performance. Finally, the feature resolution is restored to the input resolution $(H \times W)$ of the final patch expanding layer by a 4× up-sampling operation. These up-sampled features are then applied to pixel-level classification prediction by linear projection. The details of the architecture are demonstrated in the following sections.

3.2. Visual graph mamba block

In medical image segmentation, a common challenge is to both retain detailed information and provide explicit boundaries, so as to accurately distinguish and segment regions of interest from complex and cluttered backgrounds in the image. To address this, we leverage the strengths of Graph Neural Network, sparse attention, and Vision Mamba-2 to construct a hybrid hierarchical feature extraction architecture, which is a novel feature extraction unit designed for medical image semantic segmentation. This strategy enables our model to focus more effectively on aggregating essential information from the high-resolution and complex graph patches in a data-dependent manner. Specifically, unlike the traditional way of processing images quadratically, the Dynamic Graph Network (DGN) presents 2D images as graph-level structures composed of a set of linked image-patch nodes, which makes it flexible to extract irregular and intricate patterns. In addition, the sparse attention module includes window-based attention to capture local features and deformable attention to obtain more valuable informative features by selecting key and value pairs in a data-dependent way. Moreover, the Visual State Space Dual (VSSD) module is an adaptation from Mamba-2 (Dao & Gu, 2024). It is equipped with a 2D State Space Model and an eight-way multi-scanning module, capable of handling 2D image data and leveraging the selective state mechanism to filter out less important features dynamically.

As illustrated in Fig. 2(a), the VGM block comprises two parts: a Win-VGM and a Deform-VGM, which are consecutively interconnected. At the beginning of each sub-block is a Dynamic Graph Network that converts the image into a graph representation, followed by the combination of a VSSD module with Flash Window-based Attention (FWA) or Flash Deformable Attention (FDA), to capture more informative features via filtering out less important information from data-dependent features gained by prior attentions. Next, the Feed-Forward Network is made up of an Einstein Fourier Network (EFN) that performs operations in the frequency domain as a complement to the VSSD module. DeepNorm (Wang et al., 2022a) is introduced at the residual connection for confining the model update within a constant range. The operations are as follows:

$$
\begin{aligned}
Z_{w1} &= DGN(Z_{w-1}) \\
Z_{w2} &= DeepNorm(FWA(Z_{w1})) \\
Z_{w3} &= DeepNorm(VSSD(Z_{w2})) + Z_{w2} \\
Z_{w4} &= DeepNorm(EFN(Z_{w3})) + Z_{w2} \\
Z_{d1} &= DGN(Z_{w4}) \\
Z_{d2} &= DeepNorm(FDA(Z_{d1})) + Z_{w4} \\
Z_{d3} &= DeepNorm(VSSD(Z_{d2})) + Z_{d2} \\
Z_{d4} &= DeepNorm(EFN(Z_{d3})) + Z_{d3},
\end{aligned}
\tag{1}
$$

where Z_{w1}, Z_{w2}, Z_{w3}, Z_{w4} represent the output of the Win-VGM block and Z_{d1}, Z_{d2}, Z_{d3}, Z_{d4} represent the output of the Deform-VGM block, respectively. The hierarchical structure of VGM takes into account both spatial relationships and feature channels, and is capable of learning visual patterns that incorporate local and global context information at varying levels of complexity. Details of this block are presented in the following sections.

3.2.1. Visual graph structured state space duality

The modular architecture design of the visual graph state space duality module and the SSD algorithm, integrating advanced visual components, allows us to process complex medical images effectively. Specifically, the DGN module treats an image as a graph, shown in Fig. 3, which is constructed of a set of feature nodes, $V = \{v_1, v_2, \ldots, v_N\}$, generated from the previous stem module. The graph is updated by each node connecting with its nearest neighbors via the KNN algorithm (Rong et al., 2020) as shown in Eqs. (2) and (3). The operations are as follows:

$$
\begin{aligned}
x_i' &= GraphConv(x) \\
&= Update(Aggregate(Graph, W_{agg}), W_{update}) \\
&= h(x_i, g(x_i, N(x_i), W_{agg}), W_{update}),
\end{aligned}
\tag{2}
$$

$$
g(\cdot) = max(\{x_j - x_i | j \in N(x_i)\}), \tag{3}
$$

where $N(x_i)$ denotes the set of neighbor nodes of x_i. w_{agg} and w_{update} denote the learnable weights of the aggregation operation and the update operation, respectively.

The graphs are then fed to the sparse attention module to capture important information from regions with linear space complexity. For instance, in FWA, routing information is restricted to non-overlapped windows, while in FDA, the data-dependent k and v are projected onto relevant regions from deformed sampling points learned from queries via an offset generation network. The generated query, key, and value are then fed to the flash scaling dot product attention, and an IO-aware algorithm calculates the attention score. The formula of the attention is

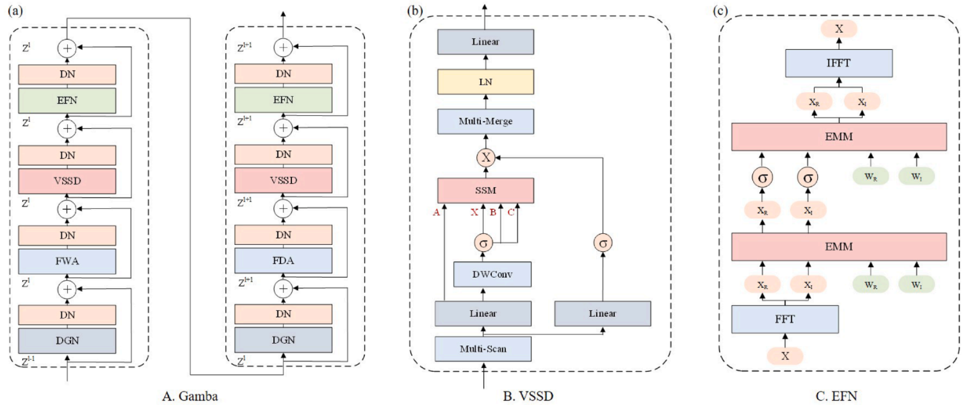

Fig. 2. (a) VGM block consists of two distinct sections, each of which comprises a consecutive series of modules: Dynamic Graph Network (DGN), Flash Window-based Attention (FWA) or Flash Deformable Attention (FDA), Visual State Space Duality (VSSD) module, and Einstein Fourier Network (EFN). A DeepNorm (DN) follows each module. (b) Visual State Space Duality (VSSD) module represents the process of instantiating parameters X, A, B, and C through multiple scanning and tensor projection parallel mechanisms. (c) Einstein Fourier Network (EFN) module represents the process of Einstein Matrix multiplication in the frequency domain for Fast Fourier transform (FFT) processing of input features.

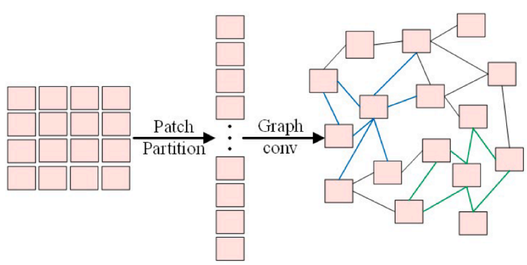

Fig. 3. Illustration of the graph representation of the nodes linked based on their contents in the image.

shown as follows:

$$z^{(m)} = \sigma\left(\frac{q^{(m)}\tilde{k}^{(m)T}}{\sqrt{d}} + \phi(\hat{B}; R)\right)\tilde{v}^{(m)}, \tag{4}$$

where $\phi(B, R)$ represents the relation position bias.

Furthermore, the output of the attentions is passed to the VSSD module, illustrated in Fig. 2(b), to capture long-range relationships and comprehensively understand the contextual information across multiple-orientation scanning within the input image, as shown in Algorithm 1. Specifically, the first step of VSSD is to adjust the dimension of the input data through a linear transformation. Then, a depth-wise convolution processes the data on each input channel separately, focusing on extracting spatial features. Subsequently, the extracted features undergo selective scanning in eight directions, including horizontal, vertical, diagonal, and diagonal lines in the forward and backward directions, through the multi-scanning function. The scanning results will be stacked together. In addition, the scanning features are passed to the tensor parallelism (TP) projection. Depending on the TP degree, the input projection matrix is split into 2, 4, or 8 shards, thereby reducing the number of reduction operations per layer from 2 to 1. This also reduces the number of parameters to a certain extent. The separated features are then fed into the SSM to calculate the hidden states. Next, the outcome states are reshaped and merged. A layer normalization is also employed to standardize these features. Finally, the features are passed to a final linear layer to form the output map. This ensures that contextual knowledge of each image patch can be obtained through a compressed hidden state computed along the corresponding scanning path, thus effectively modeling the global receptive field with linear complexity. The formula of this mechanism is shown as follows:

$$\begin{aligned} &h_i = \tilde{A}_i \odot h_{i-1} + B_i(\triangle_i \odot x_i) \\ &x_i, \triangle_i \in R^{T\times P}, \tilde{A}_i, h_{i-1}, h_i \in R^{T\times P}, B_i \in R^{T\times N}, \end{aligned} \tag{5}$$

$$\begin{aligned} &y_i = C_i h_i + D \odot x_i \\ &y_i, \in R^{T\times P}, C_i \in R^{T\times N}, D \in R^{T\times P}, \end{aligned} \tag{6}$$

where B, C, Δ are produced from the input feature x. T and P denote the sequence dimension and the head dimension, respectively. This equation maps the input x to the output y through the hidden state $h(t)$. Specifically, the VSSD obtains parameters B, C, and Δ by performing operations $B = (xW_B)^T$, $C = xW_C$, and $\Delta = softplus(xW_1W_2)$, respectively, where W_B, W_C, W_1, W_2 are the projection matrices.

This design, benefiting from graph representation learning, attention mechanisms, and state space models, enhances the modeling capability not only to capture short and long-term dependencies, but also to process scaling sequence length effectively, while avoiding extra computational cost.

3.2.2. Fourier transformation

While extending the model to a large scale, we notice its trend of becoming less effective in capturing channel information and less stable, which are the issues inherent in the Mamba model. This hinders our model from achieving high performance and effective convergence during training. The goal of the proposed EFN module, shown in Fig. 2(c), is to achieve additional modeling of channel context through frequency domain convolution, and to address the instability issues during upscaling. We also leverage the advantages of the Rayleigh energy equivalence theorem to capture small objects by selectively processing the specific energy emphasis frequency components within the input image in the frequency domain. Additionally, we obtain the specific eigenvalues required for achieving convergence through nonlinear activation, thereby making the model more stable.

Specifically, the process of the Fourier operator module comprises three components: Fourier Transformation, Frequency Matrix multiplication, and Inverse Fourier Transformation. The theorems of the Fourier transformation module begin with Eq. (7) by using the discrete Fourier transform (DFT) to perform a spectral transformation, converting the input signal into constituent frequencies:

$$F(k) = \int_D f(x)\cos(2\pi kx)dx + j\int_D f(x)\sin(2\pi kx)dx, \tag{7}$$

where k, x, and j represent the frequency variable, the spatial variable, and the imaginary unit, respectively. $f(x)$ represents an input function, and $F(k)$ represents the Fourier transformation.

Table 1
Variants of the VGM-UNet architecture.

	Output Size	VGM-UNet-T224	VGM-UNet-S224	VGM-UNet-T256
Stage1	$\frac{H}{4} \times \frac{W}{4}$	Patch Embedding↓ 4 (VGMBlock(96)×2)	Patch Embedding↓ 4 (VGMBlock(96)×2)	Patch Embedding↓ 4 (VGMBlock(96)×2)
Stage2	$\frac{H}{8} \times \frac{W}{8}$	Patch Merging↓ 2 (VGMBlock(192)×2)	Patch Merging↓ 2 (VGMBlock(192)×4)	Patch Merging↓ 2 (VGMBlock(192)×2)
Stage3	$\frac{H}{16} \times \frac{W}{16}$	Patch Merging↓ 2 (VGMBlock(384)×8)	Patch Merging↓ 2 (VGMBlock(384)×18)	Patch Merging↓ 2 (VGMBlock(384)×8)
Stage4	$\frac{H}{32} \times \frac{W}{32}$	Patch Merging↓ 2 (VGMBlock(768)×2)	Patch Merging↓ 2 (VGMBlock(768)×2)	Patch Merging↓ 2 (VGMBlock(768)×2)

Algorithm 1 Vision mamba-2 processing 2D image.

```
Input: x :∈ R^{B×D×H×W};
Output: y :∈ R^{B×D×H×W};
B, D, H, W ← x.shape
ratio ← 2.0
heads ← 4                                      ▷ [4, 8, 16, 32]
d_state ← 64
d_ssm ← d_dinner ← reatio × in_channels
headdim ← int(d_ssm / heads)
x ← conv2d(x)
x ← multi_scan(x)
d_x_proj ← 2 ∗ d_inner + 2 ∗ ngroups ∗ d_state + heads
x_proj ← []
for i ← multi_scan_directions, … , Iter do
    x_proj.append(linear(d_inner, d_x_proj))
end for
x_proj_weight ← []
for t ← x_proj, … , Iter do
    x_proj_weight.append(t.weight))
end for
x_proj_weight ← nn.Parameter(stack([
  x_proj_weight], dim = 0))
zxbcdt ← einsum('b k d l, k c d → b k c l',
  x, x_proj_weight)
z0, x0, z, xBC, dt ← zxbcdt.split()
A : (heads, N) ← Parameter
x, B, C ← xBC.split()
z ← rearrange(z, 'b k (h p) l → b l (k h) p',
  p ← headdim)
x ← rearrange(x, 'b k (h p) l → b l (k h) p',
  p ← headdim)
B ← rearrange(B, 'b k (n dst) l → b l (k n) dst',
  dst ← d_state)
C ← rearrange(C, 'b k (n dst) l → b l (k n) dst',
  dst ← d_state)
dt ← rearrange(dt, 'b k h l → b l (k h)')
dt ← softplus(dt + dt_bias)
D ← rearrange(D_init,' (h p) → h p', p ← headdim)
y ← SelectiveScan(x, dt, A, B, C, D, z, ∗∗ kwargs)
y ← cat(silu(z0) ∗ x0, y, dim ← 1)
y ← multi_merge(y)
y ← out_proj(y)
return y
```

Convolution theorem 1. *The convolution theorem indicates the equivalence of the convolution functions and the corresponding point-wise product in both the spatial domain and the frequency domain.*

Rayleigh's theorem 1. *Rayleigh's Theorem indicates that the total energy of a signal shows the equivalence between the spatial domain and its frequency domain.*

This equation of the transformation process is composed of a real part $Re(F)$ and an imaginary part $Im(F)$. The complete transformation operation, essential for identifying periodic or aperiodic patterns, is presented as $F = Re(F) + Im(F)$. In addition, the following is a convolution computation in the frequency domain through matrix multiplication, as shown below:

$$F(X * W + B) = F(X) \cdot F(W) + F(B) = \mathcal{X}\mathcal{W} + \mathcal{B}, \tag{8}$$

where F and * stand for DFT and circular convolution, respectively. W and B represent the normal weight and bias of the spatial domain, respectively. In the corresponding frequency domain, $\mathcal{W}$ and $\mathcal{B}$ are the complex weight and bias, respectively.

Here, to reduce the complexity of the Fourier transform of the convolution operations, the Einstein Matrix Multiplication is introduced to perform the calculation of mixing tokens between channels, which has an effect block diagonal matrix. By learning the interactions between spatial locations in the frequency domain, the model can precisely capture signals. The equations can be expressed as follows:

$$\begin{aligned} h^{\ell} &= \sigma\left(h^{\ell-1}\mathcal{W}^{\ell}\right)h^{0} = \mathcal{X} \\ &= \sigma\left(Re(h^{\ell})\right) + j\sigma\left(Im(h^{\ell})\right), \end{aligned} \tag{9}$$

$$\begin{aligned} Re(h^{\ell}) &= EMM\left(Re(h^{\ell-1})\mathcal{W}_r^{\ell}\right) - EMM\left(Im(h^{\ell-1})\mathcal{W}_r^{\ell}\right) + \mathcal{B} \\ Im(h^{\ell}) &= EMM\left(Re(h^{\ell-1})\mathcal{W}_r^{\ell}\right) + EMM\left(Im(h^{\ell-1})\mathcal{W}_r^{\ell}\right) + \mathcal{B}, \end{aligned} \tag{10}$$

where the final output $h^{\ell} \in R^{N*D}$ is generated from the value of ℓ_{t-1} layer through an activation function σ. This expresses that, by using Rayleigh's Theorem, it is possible to concentrate the energy of specific frequency components, thereby improving the discrimination accuracy. Non-linear activation is also utilized to handle the instability issue. Beyond enhancing the capture of signals, the method also ensures the manageability of computational complexity.

$$\begin{aligned} F(k) &= \int_D f(k)e^{j2\pi kx}\mathrm{d}f \\ &= \int_D (Re(F(k))) + jIm(F(k))e^{j2\pi ks}\mathrm{d}f. \end{aligned} \tag{11}$$

Finally, we adopt the inverse FFT to transform the output back to the original time domain and produce the overall output, as shown in Eq. (11). The process helps improve the stability and prediction accuracy of the model.

4. Experiments and results

We demonstrate that our proposed VGM-UNet is an effective architecture for segmenting medical images through comparison with state-of-the-art CNNs and Transformer models on the Synapse multi-organ dataset (Landman et al., 2015) and the ACDC dataset (Zhang, 2024). We also evaluate our proposed architecture through ablation studies under specific settings in four categories: input resolutions, the number of skip-connections, model scaling structure, and equipping adapter.

4.1. Datasets

The Synapse multi-organ dataset is composed of 30 abdominal CT scans, as well as their corresponding segmentation masks, derived from

Table 2
Comparison of segmentation performance with state-of-the-art methods on the Synapse dataset. DSC (%) and HD95 (mm) are the evaluation metrics, and DSC is also used for the individual organs. ↑↓ indicates the positive direction.

Models	Mean		Aorta	GB	KL	KR	Liver	PC	SP	SM
	DSC ↑	HD95↓								
TransUNet (Chen et al., 2021)	77.48	31.69	87.23	63.13	81.87	77.02	94.08	55.86	85.08	75.62
SwinUNet (Cao et al., 2022)	79.13	21.55	85.47	66.53	83.28	79.61	94.29	56.58	90.66	76.60
MT-UNet (Kushnure & Talbar, 2021)	78.59	26.59	87.92	64.99	81.47	77.29	93.06	59.46	87.75	76.81
CASTformer (Sun et al., 2023)	82.55	22.73	89.05	67.48	86.05	82.17	95.61	67.49	91.00	81.55
PVT-CASCADE (Titoriya & Singh, 2023)	81.06	20.23	83.01	70.59	82.23	80.37	94.08	64.43	90.10	83.69
MISSFormer (Huang et al., 2021)	81.96	18.20	86.99	68.65	85.21	82.00	94.41	65.67	91.92	80.81
MIST (Rahman et al., 2023)	86.92	11.07	89.15	74.58	93.28	92.54	94.94	72.43	92.83	87.23
RWKV-UNet (Jiang et al., 2025)	84.02	15.70	89.53	68.94	87.63	84.07	95.57	69.38	90.95	86.09
VM-UNet (Zhang et al., 2024a)	81.08	19.21	86.40	69.41	86.16	82.76	94.17	58.80	89.51	81.40
VGM-UNet-T	89.76	3.40	89.98	92.78	91.91	91.20	91.34	85.04	91.78	84.10

Table 3
Comparison of segmentation performance with state-of-the-art methods on the ACDC dataset. DSC in % is the evaluation metric.

Methods	Average	RV	Myo	LV
TransUNet (Chen et al., 2021)	89.71	88.86	84.54	95.73
Swin-UNet (Cao et al., 2022)	90.00	88.55	85.62	95.83
nnFormer (Zhou et al., 2023)	91.78	90.22	89.53	95.59
MS-UNet (Kushnure & Talbar, 2021)	87.74	85.31	84.09	93.82
PVT-CASCADE (Titoriya & Singh, 2023)	91.46	88.90	89.97	95.50
MISSFormer (Huang et al., 2021)	90.86	89.55	88.04	94.99
MIST (Rahman et al., 2023)	92.56	91.23	90.31	96.14
RWKV-UNet (Jiang et al., 2025)	92.17	90.86	88.72	96.92
VGM-UNet-T	94.86	94.78	93.37	96.42

the Medical Image Computing and Computer Assisted Intervention (MICCAI) 2015 Multi-Atlas Abdominal Organ Segmentation Challenge. A total of 3779 axial enhanced abdominal sections of 8 organs, namely the aorta, gallbladder (GB), left kidney (KL), right kidney (KR), liver, pancreas (PC), spleen (SP), and stomach (SM), were obtained from these CT scans, each of which has 85 to 198 2D slices, each at the feature resolution of 512×512.

The ACDC dataset was published in 2017 by MICCAI to advance automated diagnostic technology in the field of heart image analysis. The dataset covers multimodal MRI images, including the left ventricle (LV), right ventricle (RV), and myocardium (MYO), of various heart diseases, with corresponding labels for segmentation purposes.

We evaluate our model's performance by conducting experiments on the Synapse and ACDC datasets, choosing the average Dice Similarity Coefficient (DSC) as the metric for both, and the average 95% Hausdorff Distance (HD95) as the metric for the Synapse dataset alone. In more detail, first, DSC measures the accuracy of the model's prediction function by calculating the degree of spatial overlap between the prediction label of the model and the ground truth. A bigger value indicates higher accuracy:

$$DICE(X,Y) = \frac{2 * |X \cap Y|}{|X| + |Y|}. \tag{12}$$

Second, to exclude the unreasonable distance caused by some outlier points and maintain the overall numerical stability, the top 95% distance from small to large is generally selected as the actual Hausdorff distance, which is called 95% Hausdorff distance:

$$\begin{aligned} HD(X,Y) &= max\{\mathrm{d}XY, \mathrm{d}YX\} \\ &= max\left\{\max_{x\in X}\min_{y\in Y}\mathrm{d}(x,y), \max_{y\in Y}\min_{x\in X}\mathrm{d}(x,y)\right\}, \end{aligned} \tag{13}$$

where the X and Y represent the ground truth and prediction maps, respectively. Meanwhile, $\mathrm{d}XY$ and $\mathrm{d}YX$ denote the distances.

4.2. Experimental settings

VGM-UNet is implemented using PyTorch 2.4 framework and CUDA tools 12.4. We conduct training and evaluation of this model on an NVIDIA GeForce RTX 3090 GPU. Before feeding the images to the model, we augment both datasets through operations such as rotation, zooming, flipping, horizontal shifting, and vertical shifting, allowing the model to better process previously unseen data. The size of the input image and the window-based attention, configurable according to the input resolution, are set to 224×224 and 7, respectively. During the training stage, we adopt settings from Cheng et al. (2021) with several adjustments: The batch size is set to 4, and the epochs are set to 400. Meanwhile, we use stochastic gradient descent (SGD) as an optimization operator with a learning rate of 0.0001 and weight decay of 0.0001. We also introduce a combination of the Cross Entropy and DICE as the loss function, as shown below:

$$Loss = \alpha Loss_{Dice} + \beta Loss_{Cross}, \tag{14}$$

where α and β are two hyperparameters utilized to obtain the final loss based on the balance between the impact of $Loss_{Dice}$ and $Loss_{Cross}$. This combined loss function is adopted to ensure pixel-level accuracy and model convergence.

4.3. Implementation details

U-shaped encoder-decoder architecture, such as U-Net (Ronneberger et al., 2015), takes into account the effectiveness of hierarchical feature extraction of the multi-scale feature information of images. Empirical evidence has proved the U-shaped architecture's effectiveness in segmenting medical images. We leverage this advanced design and build a novel hybrid VGM-UNet at four scales: VGM-UNet-224-Tiny, VGM-UNet-224-Small, VGM-UNet-224-Tiny-Adapter, and VGM-UNet-256-Tiny (referred to as VGM-UNet-224-T, VGM-UNet-224-S, VGM-UNet-224-TA, VGM-UNet-256-T, respectively). An overview of the architecture of VGM-UNet-224-T is illustrated in Fig. 1, and the details are shown in Table 1.

4.4. Experiment results

4.4.1. Performance on Synapse dataset

We conduct experiments and compare the results with recent state-of-the-art models on the Synapse dataset, illustrated in Table 2 and Fig. 4. The results show that our model achieves the highest mean DSC score of 89.76 and the lowest mean HD95 of 3.40. Compared with the recent works MIST (Rahman et al., 2023), RWKV-UNet (Jiang et al., 2025), MISSFormer (Huang et al., 2021), and VM-UNet (Zhang et al., 2024a), our experiment scores see an accuracy improvement of 2.84%, 5.74%, 7.80%, and 8.68%, respectively, on the DSC evaluation indicators. At the same time, our experimental scores on the HD95 evaluation

Table 4
Comparison of computation performance with state-of-the-art methods on the Synapse dataset. DSC in % is the evaluation metric.

Methods	Backbone	Params	Flops	DSC ↑	HD95↓
TransUNet (Chen et al., 2021)	CNN + Transformer	96.07	88.91	77.48	31.69
UNet-R (Hatamizadeh et al., 2022)	CNN + Transformer	86.00	54.70	79.56	22.97
HiFormer (Heidari et al., 2023)	CNN + Transformer	25.51	8.05	80.39	14.70
Swin-UNet (Cao et al., 2022)	Transformer	27.17	6.16	79.13	21.55
CSWin-UNet (Liu et al., 2025a)	Transformer	23.57	4.72	81.12	18.86
Mamba-UNet (Wang et al., 2024b)	Mamba-1	15.47	3.53	90.77	1.13
Swin-UMamba (Liu et al., 2024a)	Attention + Mamba-1	55.05	33.53	89.40	1.68
VM-UNET-V2 (Zhang et al., 2024a)	Mamba-1	17.91	3.36	77.48	3.57
VGM-UNet-T-Adapter	Attention + Mamba2	78.27	20.16	91.20	1.29

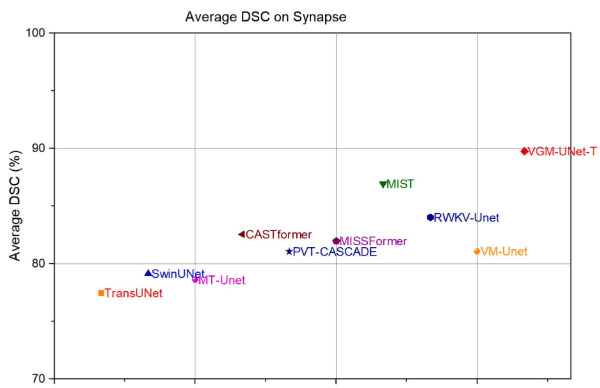

Fig. 4. Comparison of the average DSC experiment outcomes with different models on the Synapse dataset.

Table 5
Ablation study of input resolution on the Synapse dataset. DSC takes % as the evaluation index.

Model	Size	DSC	HD95	Flops
Tiny	224	89.76	3.40	20.16G
Tiny	256	89.50	1.66	26.34G

Table 6
Ablation study of scaling models on the Synapse dataset. DSC takes % as the evaluation index.

Model	Size	Skips	DSC	HD95	Flops
Tiny	224	0	88.90	2.88	20.16G
Tiny	224	1 (1/8)	89.00	2.25	20.16G
Tiny	224	3 (All)	89.76	3.40	20.16G

Table 7
Ablation study of input resolution on the Synapse dataset. DSC takes % as the evaluation index.

Model	Size	Scale	DSC	HD95	Flops
Tiny	224	2 2 8 2	89.76	3.40	20.16G
Small	224	2 4 18 2	90.70	1.66	37.48G

indicators drop 7.67, 12.30, 14.80, and 15.81, respectively. In addition, considering that among all organs in the dataset, a high level of accuracy in the pancreas segmentation task is a challenging but most pursued goal, achieving the highest score on pancreas with a significant margin proves our model's superiority in handling images of high complexity. The impressive performance can be attributed to the successful selection of advanced technologies, such as graph representation learning and the Fourier operator, and their effective integration into our hybrid encoder-decoder architecture, which successfully enhances the capture of contextual informative features.

4.4.2. Performance on ACDC dataset

The performance of VGM-UNet is also compared to recent state-of-the-art models on the ACDC dataset. The results exhibited in Table 3 and Fig. 5 show that our model outperforms most of the recent state-of-the-art works, with the highest score of 94.86 on the average DSC metric. Compared to MIST (Rahman et al., 2023) and RWKV-UNet (Jiang et al., 2025), our model improves accuracy by 2.30% and 2.69%, respectively. These results indicate that VGM-UNet achieves competitive performance, benefiting from the merits of the Visual Graph Mamba-2 module which is skilfully incorporated into our novel hybrid hierarchical feature extraction architecture. Our model is also more effective in locating small organs. The average DSC scores for segmenting the right ventricle (RV), myocardium (Myo), and left ventricle (LV) are 94.78, 93.37, and 96.42, respectively. Except for the score in the left ventricle (LV), which is lower than that achieved by RWKV-UNet (Jiang et al., 2025), our model scores better than the comparison methods. These results indicate that the hybrid architecture we employed for the model that extracts hierarchical features performs better in accurately segmenting these three organs.

4.4.3. Comparison of computational efficiency

We compare the computational performance of the VGM-UNet with other state-of-the-art models on the Synapse dataset, as shown in Table 4. Our work seeks a balance between strong modeling expressiveness and computation efficiency, besides improvements in prediction accuracy as our primary pursuit. VGM-UNet is trained from scratch, which is its unique characteristic. Here, we use floating point operations per second (FLOPs) and the parameters to measure computational complexity. The results show that the performance of VGM-UNet is superior to that of the model driven by Mamba-1, although its memory footprint is similar to that of Swin-UMamba (Liu et al., 2024a). Specifically, our model outperforms Mamba-UNet (Wang et al., 2024b), Swin-UMamba (Liu et al., 2024a), and VM-UNET-V2 (Zhang et al., 2024a) by 0.43%, 1.80%, and 13.72%, respectively. Furthermore, we conduct a qualitative experiment and compare our model's outcome with several methods, such as Swin-UNet (Cao et al., 2022), Mamba-UNet (Wang et al., 2024b), Swin-UMamba (Liu et al., 2024a), and VM-UNET-V2 (Zhang et al., 2024a), as shown in Fig. 6. These experimental results demonstrate the effectiveness of our VGM-based architecture in medical image feature extraction and representation. Compared with the existing Mamba-1-driven methods, our VGM-UNet can generate high-precision segmentation results.

4.5. Ablation studies

4.5.1. Input resolution

We conduct experiments on different scales of input resolution, including 224×224 and 256×256, to analyze the effect of input resolution on our model's modeling ability. Table 5 presents the results of training VGM-UNet on different scales of input resolution, which sug-

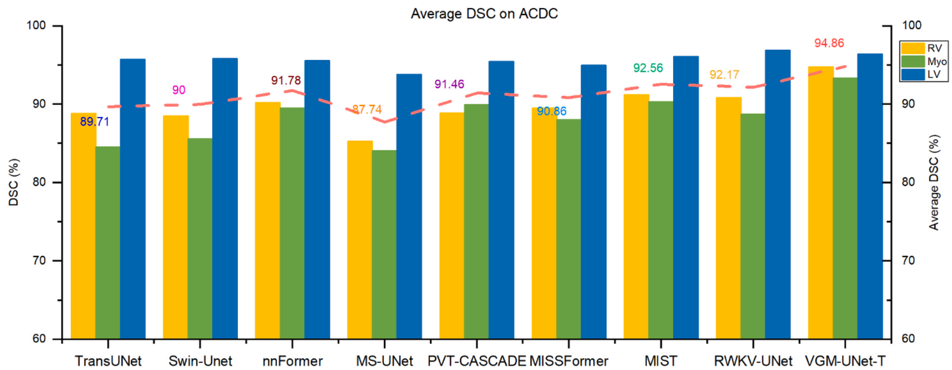

Fig. 5. Comparison of the average DSC experiment outcomes with different models on the ACDC dataset.

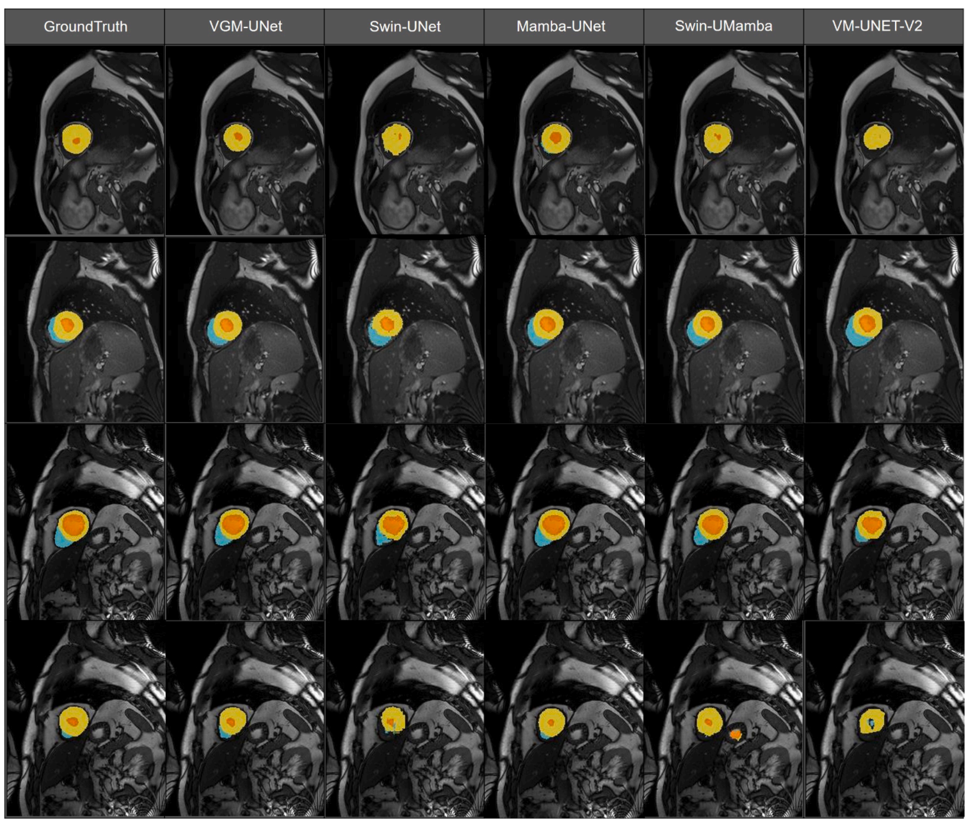

Fig. 6. Experiment outcomes compared with different methods on the ACDC dataset.

gests that input images without stark contrast in resolution lead to comparable results. Therefore, taking into account hardware requirements and computational costs, the default resolution of our model is set to 224×224, and all experimental comparisons are performed based on this setting.

4.5.2. Skip connections

Our model inherits inherent features from U-Net, such as skip connections, to preserve the critical spatial information in shallow layers, and enhance segmentation of the details of anatomical structures and pathological tissues. The resolution scales of 1/4, 1/8, and 1/16 are associated with one skip connection. To test the effect of different numbers of skip connections in the VGM-UNet, we set the number of skip connections to 0, 1, or 3, where the number 1 means keeping the connection at the scale of 1/8. Table 6 shows that an increase in the skip connection results in a better segmentation performance. Our model achieves the best precision scores when the number of skip connections is set to 3.

Table 8
Ablation study of the encoder equipped with or without adapter on the Synapse dataset. DSC takes % as the evaluation index.

Model	Size	Adapter	DSC	HD95	Flops
Tiny	224	False	89.76	3.40	20.16G
Tiny-Adapter	224	True	91.20	1.29	20.16G

Table 9
Ablation study of the effects of modules in VGM on the ACDC dataset.

Model	Size	DGN	DAT	VSSD	EFN	DSC	HD95	Flops
VGM-DGN	224		✓	✓	✓	88.80	1.60	11.50G
VGM-DA	224	✓		✓	✓	88.89	1.79	14.83G
VGM-VSSD	224	✓	✓		✓	88.61	1.41	20.14G
VGM-EFN	224	✓	✓	✓	FFN	90.41	1.44	26.18G
VGM	224	✓	✓	✓	✓	91.20	1.29	20.16G

4.5.3. Model scale

We employ “Tiny” and “Small” scaling structures in our architecture and evaluate their impacts on the performance of our model. We set the depth of the “Tiny” model to [2,2,8,2], while that of the “Small” model to [2,4,18,2], so that the models have significant contrast in layer number, which is supposed to have an impact on their performance. Table 7 demonstrates that the model’s performance improves with the increase in the layer number. Compared with the experiment on input resolution, we choose the “Tiny” model for all comparison experiments.

4.5.4. Adapter

To verify whether the model has further accuracy improvement potential, we add an adapter module (Chen et al., 2024) to the encoder before passing the input data to the VGM block. The adapter module can leverage the hierarchical feature extraction to enhance accuracy and robustness based on the non-linear operation of the Gaussian Error Linear Units function (GELU) within two linear layers. Table 8 shows the effectiveness of the add-on module, and the satisfactory result indicates our model’s great capacity for further performance improvements.

4.5.5. Effects of the modules

To make Mamba-2 (Dao & Gu, 2024) adaptive to vision data in the medical image semantic segmentation task, we introduce the SSD framework to construct a hybrid hierarchical visual backbone by consecutively connecting Graph Neural Network, sparse attention, Vision Mamba-2, and EFN via DeepNorm (Wang et al., 2022a) for feature transformation. We conduct an ablation study to evaluate the effects of these modules. We adopt the VGM-Tiny architecture and adjust the feature dimensions to compare the performance of each module by removing or replacing one of the modules. From Table 9, we can observe that when one of the modules is removed, the performance of the visual backbone will be significantly compromised. By adding more feature transformation modules (including DGN, Deformable Attention, VSSD, and EFN), a higher accuracy rate can be achieved. This experiment demonstrates the feasibility of our design.

5. Discussion

Our ultimate goal is to build an efficient medical image segmentation model adapted to various tasks in few-shot learning. Although our model achieves competitive performance, there are still existing deficiencies when performing some challenging cases. As shown in Table 7, the model’s performance improves only insignificantly when expanded to a larger scale. Although DeepNorm (Wang et al., 2022a) is used in residual connections to alleviate similarity for improving training stability when deepening the network, its performance is still not satisfactory. It may be necessary to optimize the network structure to enhance the effectiveness of residual connections, for example, by introducing hyper-connections. Additionally, the results from combining the attention mechanism and Mamba-2 (Dao & Gu, 2024) through the state space duality framework, as shown in Table 4, indicate that there is room for reducing model parameters and FLOPS. When evaluating the effectiveness of the module, we found that most of the memory usage and computational costs originated from the VSSD module. Simplifying the structure of the backbone network to render the Vision-Mmaba-2 a more lightweight network may help to manage the demand for computing resources in a controlled manner. As a remedy for this limitation, by replacing the hybrid architecture with a pure and restructured algorithm of the graph deformable vision-mamba model, fusing Graph, deformable, and Selective SSM into a basic network, and implementing this model based on structured state space duality with semiseparable matrices using a hardware-aware algorithm (Dao et al., 2022; Dao & Gu, 2024), the complexity and the number of parameters can be significantly reduced, its efficiency can be improved, and the issue of overfitting can be alleviated. Furthermore, while our model is specifically designed for semantic image segmentation tasks, it may lack the flexibility to process instance or panoptic image segmentation tasks.

6. Conclusion

In this work, we have constructed a novel Hybrid U-shaped architecture, VGM-UNet, for medical image segmentation. We have addressed the limitations of previous conventional architectures, which prevent them from achieving a high level of accuracy in visual tasks, and presented a detailed study of combining graph neural networks, attention mechanisms, and SSMs to merge the merits of these techniques. We have successfully adapted Mamba-2 as a visual model and enhanced its expressiveness by introducing the Fourier neural operator. The results of extensive experiments clearly demonstrate the effectiveness and expressiveness of our model, which benefits from its powerful modeling capability in handling medical images. Therefore, it can be widely applied to medical image segmentation tasks.

CRediT authorship contribution statement

Hang Li: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization; **Jianan Fan:** Writing – review & editing, Project administration, Methodology, Investigation, Formal analysis, Data curation; **Weidong Cai:** Writing – review & editing, Supervision, Project administration, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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